

Mohamed Yusuf Kathree

✉ 4253340@myuwc.ac.za ☎ 079 239 7141 📍 Cape Town, South Africa 🔗 [linkedin.com/in/mohamed-kathree](https://www.linkedin.com/in/mohamed-kathree)
🐙 github.com/Mohamed-Kathree

Professional Summary

BSc Honours Computer Science candidate at the University of the Western Cape, graduating 2026. Strong foundation in data structures, algorithms, object-oriented design, and the software development lifecycle (SDLC), with hands-on Python, Java, and C across distributed systems, machine learning, and embedded projects. Built and shipped multiple end-to-end systems including an RL-guided autonomous aircraft (PPO, JSBSim), an automated trading platform (2.5 Sharpe ratio), and a UDP-based networked security system on Raspberry Pi. Comfortable with Git version control, agile workflows, CI/CD practices, debugging complex systems, and using AI-powered tools to accelerate development. Seeking a Software Development Engineer role to design and operate scalable, fault tolerant solutions that deliver direct customer impact.

Technical Skills

Languages: Python, Java, C, C++, SQL, TypeScript (basics), ARM Assembly
Core CS: Data Structures, Algorithms, Object-Oriented Design, Complexity Analysis, Recursion
Engineering Practice: Agile, CI/CD, SDLC, Code Reviews, Unit Testing, Debugging, Git Version Control
Distributed & Cloud: Distributed Systems, Microservices (basics), REST APIs, UDP/TCP Networking, Apache Kafka, AWS (learning)
Databases: MySQL, MariaDB, SQLite3, NoSQL (basics)
Web & AI: React, Next.js, HTML/CSS, NumPy, pandas, Matplotlib, Machine Learning, Reinforcement Learning
Tools & Embedded: Linux, Jupyter, Wireshark, GNS3, Raspberry Pi, Arduino, Embedded Systems
Soft Skills: Problem-Solving, Customer Focus, Ownership, Team Leadership, Communication, Collaboration

Education

BSc Honours — Computer Science 2026

University of the Western Cape

Honours thesis: RL-Guided Fixed-Wing Glider autonomous return-to-launch system using Gymnasium PPO with JSBSim 6-DOF physics engine. Achieved 91% for Semester 1 module submission (COS731).

BSc — Computer Science 2022 – 2025

University of the Western Cape

- CS modules: COS101 (Python & Java), CSC211 (Java), CSC212 (Assembly), CSC311 (C), CSC312, Computer Networks (Java), AI (Java) · Mathematics: MAT312, MAT322 · Physics: Classical Mechanics, Electromagnetism, Quantum Mechanics.
- Strong module performance: Computer Science 311, Computer Science 312, Mathematics 312 and Mathematics 322.
- CS concepts: OOP, recursion, SDLC, CI/CD, advanced data structures, complexity analysis, system architecture, OS & concurrency, optimisation, network protocols, search & heuristics, logic, databases, software engineering, machine learning.
- Mathematics & physics concepts: single & multivariable calculus, differential equations, linear algebra (transformations, eigenvalues), abstract algebra (group & ring theory), vector calculus, classical mechanics, harmonic motion, fluid dynamics & viscosity, electrostatics & Coulomb's law, capacitance & RC circuits, quantum state & superposition.

National Senior Certificate (Bachelor's Pass) 2021 – 2021

Star College Bridgetown

Strong matric pass in Mathematics, Physical Sciences, and Life Sciences, providing a quantitative foundation for tertiary study.

Experience

Senior Student Assistant 01/2022 – 01/2025

University of the Western Cape

Faculty of Natural Science

- Promoted to Senior Student Assistant after one year of service; held the role for 3 academic years, demonstrating ownership and reliability.
- Supported undergraduate students across multiple science modules.
- Collaborated with faculty to set up computer laboratories and resolve technical issues for 30+ students per day; managed administrative duties and documentation for the Faculty of Natural Science.

Team Member 11/2024 – 12/2024

YES Summit 2025

Collaborated in a cross-disciplinary team to design an autonomous-drone concept for mining operations; team awarded delegate passes to the Mining Indaba 2025 conference for the winning proposal.

Projects

RL-Guided Fixed-Wing Glider (Return-to-Launch)

Ongoing

- Python, JSBSim, Gymnasium (PPO), Reinforcement Learning
- Designed and implemented a simulation-first reinforcement learning system in Python to autonomously navigate a hand-launched fixed-wing glider back to launch.

- Architected an end-to-end ML training pipeline using Gymnasium PPO with four-stage curriculum and domain randomisation; integrated JSBSim for validated 6-DOF aerodynamics; debugged reward shaping iteratively.
- Achieved 28–29% return-to-launch success under zero-wind baseline conditions.

Automated CFD Trading Algorithm

Ongoing

- Python, Quantitative Analysis, Financial Modelling
- Engineered and backtested a fully automated trading system in Python, achieving a 2.5 Sharpe ratio on 2025 FY data; applied data structures and statistical algorithms for market data processing, signal generation, and execution; instrumented logging and metrics.

F1 Database Management GUI

2025

- Python, MySQL, MySQL Workbench, SQL
- Designed and built a Formula 1-themed desktop application connecting a Python front-end to a MySQL database; implemented predefined and custom SQL queries, CSV export, and structured logging. Applied object-oriented design, database normalisation, Git version control, and peer code reviews.

Peer-Review & Tutoring Platform

2025

- React, Next.js, HTML/CSS
- Developing a full-stack web application with React and Next.js for peer reviews and tutoring sessions, including a real-time chat-room module; applied component-based architecture, responsive design, and RESTful API patterns.

Networked Security System

2025

- Raspberry Pi, UDP Networking, Embedded Systems, Java
- Led a software engineering team as Team Captain to ship a distributed alarm system: a Raspberry Pi node transmits UDP signals to a central server, which broadcasts alerts to all connected clients. Implemented fault tolerant client registration, non-blocking I/O, persistent logging, and coordinated agile workflow with code reviews.

Physics & Mathematical Modelling Simulations

2022 – 2023

2022–2023

- VPython, Python, Scientific Computing
- Built computational models in VPython to simulate simple/damped harmonic motion, projectile motion with drag, viscous fluid resistance, and charged-particle dynamics; applied numerical integration (Euler method) and validated outputs against analytical solutions.

Awards & Achievements

3rd Place — Cybersecurity CTF Challenge

Competed against national and international universities.

2.5 Sharpe Ratio

Automated trading algorithm backtested on 2025 financial year data.

Certifications

Beginning C++ Programming: Online Course (Completed)

Forex Trading A-Z with Live Examples of Real Trades: Online Course (Completed)

Advanced Technical Analysis Trading Strategies: Online Course (Completed)

Technical Analysis & Swing Trading: Online Course (Completed)

Electronics, PCB Design, Arduino, and Microcontroller Development: Crash Courses (Completed)

Data Science and Machine Learning: Self-Study / Online Resources (Ongoing)

Languages

English: C2 — Proficient

Afrikaans: B1

References

Nathan Lyle Naidoo, Supervisor & Lecturer, UWC
nlnaidoo@uwc.ac.za, 084 805 2048

Andre Henney, Lecturer, UWC
ahenney@uwc.ac.za, 083 532 9141

Candice Hendricks, UWC Faculty of Natural Science
canhendricks@uwc.ac.za, 076 142 1364